

Ethernet Security

Module 24

Duration: 34:12

Video: <https://www.youtube.com/watch?v=uaDXhq0k5sc>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Auto-generated YouTube transcript for personal study. Spoken wording belongs to the original lecture. Follow the video link above; this file is not an official CEC/SWAYAM publication.

Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana I welcome you all to the video lecture series of cyber security information security in this lecture we will study about ethernet security which is basically used to secure wired local area network and metropolitan area network first of all please see the contents which I shall cover after covering brief introduction about ethernet security I shall take up 1 minute, 1 second ethernet threats which will be followed by ethernet Security Solutions in last I shall take up Max SEC protocol for ethernet security so now let us start ethernet has its roots in the 1970s in a radio experiment called Aloha

net developed under the direction of Norman Abramson at the University of Hawaii at about same time gerox Corporation created a landan with a data rate of 3 Mbps using a carrier sense multiple assess with Collision detection protocol csma by CD in 1980 the 10 Mbps ethernet version 1.0 spe specification was jointly released by digital Equipment Corporation Intel Corporation and Grox Corporation this was picked up by i e and became the i e 802.3 standard in 1983 a chunk of data transmitted by ethernet over the wire is called a frame on an Ethernet Network only a single node should be transmitting a frame at any time if multiple systems are transmitting simultaneously a collision will occur which can cause both signals to fail and require the systems to transmit

their frames again these properties are summarized under ethernet designation as a carrier sense multiple assess with Collision detection CSM by CD protocol figure shows a wired network using ethernet ethernet supports both optical fiber and copper cabling yellow color infigure shows optical fiber and blue color shows twisted pair care CES in figure we can see Server form and pces connected to fast ethernet 100 Mbps network using switches the ethernet standards comprise several wiring and signaling variants of the OSI physical layer in use with ethernet the original 10 base 5 ethernet uses coxl cable as a shared medium while the newer ethernet variant use twisted pair and fiber optic links in

conjunction with hubs or switches over the course of its history ethernet data transfer rates have been increased from the original 10 megabits per second to the latest 100 gbits per second with 400 gabs per second expected by early 2017 10 MVPs speed known as ethernet 100 Mbps known as fast ethernet 1,000 Mbps known as gigabit Ethernet 10,000 Mbps speed as 10g ethernet and 100,000 Mbps as 100g ethernet 400 GB speed is also expected in 2017 ethernet is the only survivor of the wired Lan War it is hard to find an IP packet that has not passed over an Ethernet segment one important reason for this is ethernet's Simplicity and ease of configuration however ethernet has always been known to be an insecure technology recent successful malware

attacks and the move toward cloud computing in data center demand that attention should be paid to the security aspect of ethernet now we shall move on to known ethernet related security threats the key historical reason for the security vulnerabilities of ethernet is that security has never been a major consideration in its design the basis for attack is gaining assess to the Target ethernet segment the attackers may be an Insider with full assess right or may have found an ethernet connection in a public public space before we describe the most prominent methods for attacking ethernet segment let us see how an attacker May utilize the network assess techniques used by attackers are learning about the private Network

topology and the network traffic for use in attack gaining control over switch is routers or servers in the landan eaves dropping manipulating information disrupting the availability of the network the most prominent methods for attacking ethernet segments are network and system assess traffic confidentiality traffic Integrity denial of service system Security First of all let us see network and system assess type of attacks assess to the network is a prerequisite for attacks and a necessity for all types of attackers assess can be achieved by connecting equipment to the network or 8 minutes, 1 second by gaining control of existing resources various methods adopted for gaining unauthorized assess to network and system are unauthorized

joins connect to an Ethernet segment by gaining assess to an unconnected port on a switch it can be done by gaining physical assess to the switch or gaining assess to a wall socket or removing the cable from a computer and plugging it into another computer or plugging in a switch between the existing computer and the socket unauthorized expansion of the network the architecture of the ethernet allows users to access spend the network 9 minutes, 1 second by installing their own switches or wireless access points which in turn allows other people join the network vland join if a switch listens for vand management protocol on host port a host can act as a switch and join all vlans on some switches

the ports can be configured to not transmit VLAN management protocols but they will still listen for these protocols an attacker can probe the switch for hidden features like this vand tagging and hoping an attacker can create ethernet frames that have a ven tag and thus inject frames to V lanss to which they are not supposed to have access remote access to the landan

access to an Ethernet segment can be achieved by gaining higher layer access to a host on the segment for example by using social engineering and to get a user at the Target Network to open a remote system administration service which then connects to a host on the internet and enables the attackers to access the ethernet layer switch control as previously mentioned switches

are shipped with default or no passwords and the password can usually be physically reset if an attacker gains control of a switch traffic can be rerouted by switching links down claiming the STP route by Raising the priority of the switch or dose selected links second type of attack methods are traffic confidentiality traffic on the network can be useful in itself and also serve the attackers in search of Targets in such type of attack an attacker not only gains information being transmitted but also gets authentication information like passwords and network topology information that can be used for further purposes third type of attacks are of traffic Integrity type it is done by modifying traffic on

the network for example an attacker can imitate A bank's web server to a user and imitate the user to the bank server and gain temporary control of the users's bank account different ways to modify traffic on the network are ARP and DHCP poisoning host capture traffic intended for another host post just by sending an ARP message to the sender with the intended receiver's IP address and the attacker's MAC address in a similar way a host can detect broadcast DHCP server request and raise the server to reply them first upon success the attackers can assign a Gateway and DNS server to the Target host along with its IP address and control the host traffic at will man in the middle if an attacker can direct traffic to pass through his node and that

traffic is not protected by an Integrity verification mechanism the attacker can easily modify the traffic such attackers are called man in the middle attack session hijacking ethernet is a stateless protocol but many higher level protocols create a session once a session is set up it is often assumed to be trusted and no further traffic verification is made if an attacker can Eaves drop on or otherwise gain enough information about an session the attacker can recreate the session and act like one end point next type of ethernet attacks are denial of service attack the attacker's motivation for dose attacks is not to gain access to data but to prevent its use the attacks can cause total loss of service or

degradation of service a dose attack can be performed in various ways resource exhaust attack resource exhaust attack can Target the control and management planes of a switch by sending frames that require additional processing and handling protocol based dose the STP that makes a tree out of a mesh network is designed to be self-configuring an attacker that controls a node on the network can send STP messages and pretend to be a switch the whole switching Network can be brought to halt by flooding it with STP tcns or other STP control messages is next we shall see solutions to Common ethernet attacks the security of ethernet has been improved by standard organizations equipment vendors and the

research Community traditionally ethernet lack of security has been solved by defining any ethernet segment as unsecure and require iring it to be placed inside a protected domain like placing it behind a firewall in a secure building with trusted staff higher layer cryptographic Solutions like IPC and TLs are used to solve the remaining issues cryptography carries its own cost and thus cannot be considered a universal Sol solution now let us see few existing Security Solutions first of all let us see router based security methods in this Solution One Central ethernet switch is replaced with an IP router the IP router partitions ethernet Network into several segments each new segment is a separate

broadcast domain ARP STP VLAN and Mac address table based attacks are no longer possible between the segments inside the segment the same attacks remain feasible unless each switch is replaced with a multi-port router replacing a switch with a router will incur some cost an IP router requires configuration such as address allocation and default route configuration occasionally this can be automated as is the case for residential Broadband where the topology is clear to the assess router the router also prohibits easy Mobility a host may move in the ethernet Network nwor and keep its IP and Mac 18 minutes, 1 second addresses as it is the MAC address tables in switches are updated automatically compared to an ethernet switch an IP router provides a considerable amount of protection

against other users connected to the same router second method is assess control method an attacker needs assess before being able to perform any attack untrusted entities can be kept out by limiting assess to the network or requiring authentication few methods of protection using assess control mechanism are physical protection of the network network equipment can be located in locked cabinets and rs and wiring installed inside walls to prevent unauthorized assess segmentation and V Lans inside ethernet the i e 802.1q virtual L mechanism provides a way to limit broadcast and other traffic to specific segments vlans are logically separate within the same physical installation and Define security domains inside one network individual

vlans each host on the ethernet can also be placed into its own VLAN using i e 82.1 a q inq double tagging or vendor provided private vand switch configuration the PV Lan technique uses switch configuration to isolate host and only let their traffic pass to one promiscuous port typically connected to a router and to the internet each host sees only itself and the host connected to the promiscuous port and only a few VLAN IDs are needed at the trunks to indicate PV Lan traffic assess control list assess control list security feature is not part of the ethernet specification switch vendors have added various types of capabilities by themselves the ethernet frame does not have many features for a simple ethernet

frame ACL the usable attribute are the senders or receiver's Mac address or The Ether type field assess can be limited based on Mac addresses but several service specific ACLS are commonly implemented authentication based assess control I E 802.1x port authentication supports several types of authentication credentials such as a username and password pair or

a certificate and corresponding private key in i.e. 802.1x the client communicates with the authentication switch and the switch verifies the authentication from the database in authentication server as shown in figure 802.1x uses extensible Authentication Protocol EAP that has broad support for different types of authentication methods and structures 802.1x authenticates a host at the

beginning of a session attaching the MAC address to a specific port in the switch so this these were access control based Security Solutions which limit the availability of targets to attackers now we shall see secure protocols for ethernet security the targets can also be made harder to reach by adding security features to protocols and such security Protocols are implemented by following methods encryption and integrity verification cryptography can solve integrity and confidentiality requirements i.e. 802.1 AE Max SEC forms encrypted connections between host and switches protecting confidentiality and integrity of the content in the frame deployment requires software installation and configuring authentication for each participating

Network entity maxc provides protection against Intruders to the network preventing reading and modification of data frames we will see details of Max protocol later in this lecture securing address resolution protocol ARP creates a major vulnerability in the ethernet architecture information gained by DHCP snooping can be used to prevent ARP spoofing attacks by attaching Mac addresses to their corresponding IP addresses and ports based on information gained from DHCP messages DHCP snooping suffers from a potential lack of scope as a single switch cannot see the allocations made to host whose path to the DHCP server does not pass through this switch last type of security methods are security monitoring based solution

in previous sections security techniques that are mostly proactive and once set do not require active participation from external systems or human interaction were described active Technologies enhance the protection of the network by implementing security monitoring based Solutions which are as explained below ethernet firewall and deep packet inspection firewalls are used to limit traffic between Network segments and can be considered more complex cases of access control list including stateful features current firewall products can operate on all Network layers intrusion detection and prevention 26 minutes, 1 second systems intrusion detection system IDS and intrusion prevention system IPS use

DPI to identify Network attacks usually from a signature library of known attacks such systems are rarely useful for ethernet Network protection planning configuration and Administration good Network Administration and practices can considerably influence the aspect of an Ethernet Network several of the Technical Solutions mentioned previously require configuration and constant adjustment at the network topology changes so these were techniques to secure ethernet Network now let us move 27 minutes, 1 second on to last part of this lecture that is maxc protocol Max SEC that is Max security defined by i.e. h2.1 AE standard allows authorized systems that attach to and interconnect lens in a network they

maintain confidentiality of transmitted data and

to take measures against frames transmitted or Modified by unauthorized devices maxc protocol facilitates following maintenance of correct network connectivity and services isolation of denial of service attacks localization of any source of network communication to the land of 28 minutes, 1 second origin construction of public networks offering service to unrelated or possibly mutually suspicious customers using shared land infrastructures secure communication between organizations using a landan for transmission incremental and non disruptive deployment protecting the most vulnerable Network components maxc operates in data part of ethernet frame as shown in figure destination address and Source

address are same in mpdu or data part of ethernet frame SEC tag is inserted before data and I see V is inserted after data Max SEC comprises modification and additions to the max service data unit msdu conveyed by each frame transmitted by a user of the protocol the Mac security tag SE tag conveys parameters that identify the protocol and identify the key to be used to validate the received frame and provide replay protection these parameters are TAG control information TCI Association number a packet number PN and secure Channel identifier SCI the secure data field conveys the US user data encrypted if confidentiality is provided the icv ensures the Integrity of the Mac destination address Mac Source address SE tag and user

data figure illustrates the transmission and reception of a frame by maxc on transmission side the frame name is first assigned to an security Association sa identified locally by its Association number the a is used to identify the security Association key s AK and the next packet number PN for that sa a the a sci and the PN encoded in the SE tag are given to protect unone transmitter side protect unit on transmitter side transmits icv and user secure data to receiver on reip of mexc frame the a sci and PN are extracted from the SE tag at receiver side the A and seci are used to assign the frame to an sa and hence to identify the security Association key s a k the validation function on receive

side of the current Cipher suit is presented with the sack pnci the destination and source address of the frame together with the octets of the SE tag secure data and icv if the Integrity of the frame has been preserved and user data can be successfully decoded from the secure data a valid indication and the octets of the user data are returned if the receive frame is Val replay protection is applied by checking that the received PN is not less than the lowest acceptable PN for the sa if the check succeeds the parameter of the frame unchanged from those transmitted are presented to the max SEC receiver and lowest acceptable PN is updated and hence transmission is successfully accepted so friends this was about

ethernet security finally summarizing we learned about introduction to ethernet security which was followed by ethernet security threats and then ethernet Security Solutions in the end we discussed working of very famous Max sex security protocol hope the contents explained in this lecture were understandable and helpful hope to see you again in next

lecture till then goodbye enjoy the day thank you